Northwestern University
CIERA: Center for Interdiscplinary
Exploration and Research in Astrophysics
1800 Sherman Ave., Evanston, IL 60201
https://jamesjohnson.space

James W. Johnson

Curriculum Vitae

jjohnson10@carnegiescience.edu

Academic Positions

Northwestern University — Evanston, Illinois

2026 – present **Postdoctoral Researcher**
Center for Interdisplinary Exploration and Research in Astrophysics (CIERA)
LSSTC Data Science Fellowship Program Coordinator
Supervisor: Prof. Adam A. Miller

Carnegie Institution of Washington – The Observatories — Pasadena, California

2023 – 2026 **Postdoctoral Fellow**, Carnegie Theoretical Astrophysics Center (CTAC)
Supervisor: Dr. Ana Bonaca

Education

The Ohio State University — Columbus, Ohio

July 2023 **Ph.D. in Astrophysics**, Dissertation Advisor: Prof. David H. Weinberg
From Dwarfs to Spirals: Chemical Evolution of Galaxies across Stellar Mass and the Implications for Nucleosynthesis

November 2019 **M.S. in Astrophysics**

Vanderbilt University — Nashville, Tennessee

May 2017 **B.A.**, Physics (major), Astronomy (minor), *cum laude*
Highest Honors in Astronomy, Thesis Advisor: Prof. Andreas A. Berlind

Research

39	14	25	1,100+	19
Journal Publications	**1st & 2nd Author**	**Contributing Author**	**Citations**	**H-Index**

Interests

Galactic chemical evolution – The Milky Way – Dwarf galaxies – The astrophysical origin of the elements – Big bang nucleosynthesis – Near field cosmology – Astronomical software

NASA ADS Libraries (A full list of my journal publications is included.)

All My Papers https://ui.adsabs.harvard.edu/public-libraries/rIqfpNKmSdaOMIAhkk2VzQ
1st & 2nd Author https://ui.adsabs.harvard.edu/public-libraries/go1WSseGTMeft2SxdESAgw
Co-Author https://ui.adsabs.harvard.edu/public-libraries/sZkjSf_XRSKSRykqBe6B_w

Seminars & Conference Presentations

Invited Talk **APOGEE's Galactic Odyssey** 2026
Centro de Estudios de Fisica del Cosmos de Aragón (Teruel, Spain)
Invited Talk **Galactic Metal Cycles**, University of Notre Dame (Kylemore Abbey, Ireland) 2026

Invited Seminar	**University of California, Davis**, Dept. of Physics and Astronomy (Davis, CA)	2025
Invited Seminar	**Stockholms Universitet**, Dept. of Astronomy (Stockholm, Sweden)	2025
Invited Seminar	**Uppsala Universitet**, Dept. of Physics & Astronomy (Uppsala, Sweden)	2025
Poster	**Small Galaxies, Cosmic Questions - II** University of Durham (Durham, United Kingdom)	2024
Invited Talk	**DHWFEST: Dark, Hot, Warm, and Fuzzy mattEr in Space and Time** University of Utah (Salt Lake City, UT)	2024
Invited Seminar	**Lund University**, Dept. of Physics (Lund, Sweden)	2024
Contributed Talk	**ADONIS: Abundance Gradients in the Local Universe** Munich Institute for Astro-, Particle, and BioPhysics (MIAPbP) (Munich, Germany)	2024
Contributed Talk	**Surveying the Milky Way: The Universe in Our Own Backyard** California Institute of Technology (Pasadena, CA)	2023
Dissertation Talk	**241st American Astronomical Society Meeting** (Seattle, WA)	2023
Contributed Talk	**GALactic Archaeology with Hermes (GALAH) Science Meeting** (virtual)	2021
Poster	**236th American Astronomical Society Meeting** (virtual)	2020
Invited Seminar	**University of California, Santa Cruz** (Santa Cruz, CA)	2019
	Sloan Digital Sky Survey (SDSS) Collaboration Meetings	
Contributed Talk	Katholieke Universiteit Leuven (Leuven, Belgium)	2026
Contributed Talk	Max Planck Institut für Astronomie (Heidelberg, Germany)	2025
Contributed Talk	New Mexico State University (Las Cruces, NM)	2024
Contributed Talk	Virtual	2021
Contributed Talk	Virtual	2020

Astrophysical Software Development

Versatile Integrator for Chemical Evolution (VICE)
Lead developer and license owner (Spring 2018 – Present)
Documentation: https://vice-astro.readthedocs.io
Source Code: https://github.com/giganano/VICE.git
Install: https://pypi.org/project/vice

Observing Programs

PI: *The First Extragalactic Measure of the Helium Isotopic Ratio – A New Test of Fundamental Physics*

2024B	WINERED spectrograph, 18 hours (Clay 6.5-m Telescope, Las Campanas Observatory)
2025A	MIKE spectrograph, 6 hours (Clay 6.5-m Telescope, Las Campanas Observatory)

Mentorship

Cal-Bridge Summer Research Program

2025 – Present	**Christopher Giudice** (undergraduate), San Francisco State University Project: The Chemical Equilibration Timescale of the Milky Way Disk Now: M.Sc. student at California State University, Long Beach
2024 – 2025	**Damien Tessmer** (undergraduate), San Diego State University Project: Identifying trends in stellar nucleosynthesis with SDSS-V data

The Ohio State University

2021 – Present	**Liam O. Dubay** (graduate), Dept. of Astronomy Projects: Galactic chemical evolution in the Milky Way (arxiv:2404.08509, 2508.00988, 2608.26261)
2022 – Present	**Daniel A. Boyea** (undergraduate), Dept. of Astronomy Project: Investigating the astrophysical origin of carbon Now: Ph.D. student at University of Groningen (Groningen, Netherlands)

2021 – 2023 **Miqaela K. Weller** (graduate), Dept. of Astronomy
Projects: Investigating the astrophysical origin of helium (arxiv:2404.08765)

2022 – 2023 **Lindsey Stultz** (undergraduate), Dept. of Physics
Polaris Near-Peer Mentorship Program

Honors & Awards

2023 **CTAC Postdoctoral Fellowship**, Carnegie Science

2022 **Ann S. Tuttle Paper Prize**, Ohio State, Dept. of Astronomy
Annual award to the top graduate student-led publication of the previous year
Paper: Johnson J.W., et al., 2021, MNRAS, 508, 4484 (arxiv:2103.09838)

2022 – 2023 **Presidential Fellowship**, Ohio State, College of Arts & Sciences
Financial support for final-year graduate students

2017 – 2018 **University Fellowship**, Ohio State, College of Arts & Sciences
Financial support for first-year graduate students

2017 **Larry Ross Cathey Award**, Vanderbilt, Dept. of Physics & Astronomy
Outstanding graduating senior studying astronomy

Inducted 2015 **Sigma Pi Sigma Physics National Honor Society**, Vanderbilt Chapter

7 of 8 semesters **Dean's List**, Vanderbilt, College of Arts & Sciences

Community Involvement

Carnegie Science Observatories

2024 – Present **Research Mentor**, Cal-Bridge Summer Research Program (2 students)

2024 **Advancing Inclusive Mentoring**
12+ hours of instruction and discussion on equitable mentorship practices

Polaris Near-Peer Mentorship Program

Graduate student-leg organization at Ohio State dedicated to fostering a more inclusive environment and improving retention of underrepresented minority groups in the Dept. of Physics and the Dept. of Astronomy. Website: https://u.osu.edu/polaris.

2022 – 2023 **Leadership Committee** (budget: ~$60,000/year)

2022 **Academic Facilitator**, Undergraduate Residential Summer Access Program
Early-arrival program for first-year undergraduate students

2022 – 2023 **Near-Peer Mentor** (1 student)

Teaching

Undergraduate Python Coding Workshop

Program Creator, six sessions, ~25 hours of instruction and exercises
Website: https://jamesjohnson.space/bootcamp
Source material: https://github.com/giganano/PythonBootcamp

2020 – 2023 **Full program (annually)**: Summer undergraduate research students
The Ohio State University, Dept. of Astronomy

2022 **Full program**: 1^{st}- & 2^{nd}-year graduate students
The Ohio State University, Dept. of Astronomy

2024 **Select sessions**: CASSI & Cal-Bridge undergraduate research students
Carnegie Science Observatories

The Ohio State University, Department of Astronomy: Graduate Teaching Assistant

2018 – 2020	**Astronomy 1101: From Planets to Cosmos**	5 sections
2019	**Astronomy 1142: Black Holes**	1 section
2019	**Astronomy 1221: Astronomy Data Analysis**	1 section
2018	**Astronomy 1140: Planets and the Solar System**	1 section

Miscellaneous

2022 – Present	**Manuscript Referee**: ApJ, MNRAS, PASJ, A&A
2024 – 2026	**Working Group Co-Chair**, Galactic Genesis, Sloan Digital Sky Survey-V
2024 – 2026	**"Morning Tea" co-organizer** (daily arXiv discussion), Carnegie Science
2024 – 2025	**External Panelist**, Hubble Space Telescope Proposal Review, Cycles 32 & 33
2021 – 2023	**"Galaxy Hour" meeting organizer**, Ohio State, Dept. of Astronomy
2017 – 2023	**Diversity Journal Club**, Ohio State, Dept. of Astronomy
June 2020	**Real Scientists Germany Online Outreach** Blog: https://tinyurl.com/jamesjohnsonrealscientistsDE Twitter: https://twitter.com/realsci_DE
2015 – 2017	**Undergraduate Tutor, Proctor, Grader** Vanderbilt University, Dept. of Physics & Astronomy
2015	**Cosmic Ray Observatory Project**, Instrumentation lab University of Nebraska-Lincoln, Dept. of Physics

Journal Publications

First & Second Author (reverse chronological order)

1. *Towards a measurement of the primordial helium isotope ratio*
   Cooke R.J., **Johnson J.W.**, Noterdaeme P, Pettini M., Welsh L., Wong A., Peroux C.
   2026, ApJ, 1003, 225 – 240 arxiv:2605.00122

2. *The galactic chemical evolution of carbon: Implications for stellar nucleosynthesis*
   Boyea D.A., **Johnson J.W.**, Weinberg D.H.
   2025, MNRAS, 550, 1155 – 1174 arxiv:2511.20752

3. *Metals versus Non-Metals: Chemical Evolution of Hydrogen and Helium Isotopes in the Milky Way*
   Johnson J.W., Weller M.K., Cooke R.J.
   2025, submitted to AAS Journals, under peer review arxiv:2510.08689

4. *That's so Retro: The Gaia-Sausage-Enceladus Merger Trajectory as the Origin of the Chemical Abundance Bimodality in the Milky Way Disk*
   Johnson J.W., Feuillet D.K., Bonaca A., de Brito Silva D.
   2025, submitted to AAS Journals, under peer review arxiv:2510.08688

5. *Constraints on Radial Gas Flows in the Milky Way Disk Revealed by Large Stellar Age Catalogs*
   Johnson J.W.
   2025, submitted to AAS Journals, under peer review arxiv:2510.05223

6. *A Galactic Perspective on the (Unremarkable) Relative Refactory Depletion Observed in the Sun*

Rampalli R., **Johnson J.W.**, Ness M.K., Edwards G.H., Newton E.R., Griffith E.J., Bedell M., Wang K.
2025, ApJ, 1002, 127 – 147 arxiv:2509.03577

7. *Rising from the Ashes II: The Bar-driven Abundance Bimodality in the Milky Way*
Beane A., **Johnson J.W.**, Semenov V., Hernquist L., Chandra V., Conroy C.
2024, ApJ, 985, 221 – 233 arxiv:2410.21580

8. *The Milky Way Radial Metallicity Gradient as an Equilibrium Phenomenon: Why Old Stars are Metal-Rich*
Johnson J.W., et al.
2024, ApJ, 988, 8 – 35 arxiv:2410.13256

9. *Dwarf galaxy archaeology from chemical abundances and star formation histories*
Johnson J.W., et al.
2023, MNRAS, 526, 5084 – 5109 arxiv:2210.01816

10. *Binaries drive high Type Ia supernova rates in dwarf galaxies*
Johnson J.W., Kochanek C.S., Stanek K.Z.
2023, MNRAS, 526, 5911 – 5918 arxiv:2210.01818

11. *Empirical constraints on the nucleosynthesis of nitrogen*
Johnson J.W., Weinberg D.H., Vincenzo F., Bird J.C., Griffith E.J.
2023, MNRAS, 520, 782 – 803 arxiv:2202.04666

12. *Stellar migration and chemical enrichment in the Milky Way disc: a hybrid model*
Johnson J.W., et al.
2021, MNRAS, 508, 4484 – 4511 arxiv:2103.09838

13. *The impact of starbursts on element abundance ratios*
Johnson J.W., Weinberg D.H.
2020, MNRAS, 498, 1364 – 1381 arxiv:1911.02598

14. *The secondary spin bias of dark matter haloes*
Johnson J.W., Maller A.H., Berlind A.A., Sinha M., Holley-Bockelmann J.K.
2019, MNRAS, 486, 1156 – 1166 arxiv:1812.02206

Contributing Author (reverse chronological order)

1. *A Broken Clock is Right Twice a Day: [Ce/Mg] is not a Universal Chemical Clock*
Dubay L.O., et al., incl. **Johnson J.W.**
2026, submitted to AAS Journals, under peer review arxiv:2608.26261

2. *The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper*
SDSS Collaboration, et al., incl. **Johnson J.W.**
2026, submitted to AAS Journals, under peer review arxiv:2607.26149

3. *Milky Way Mapper Decoded Abundances II: From patterns to paths*
Ness M.K., et al., incl. **Johnson J.W.**
2026, MNRAS, 550, 960 – 976 arxiv:2605.21735

4. *Milky Way Mapper Decoded Abundances I: Shared disk enrichment patterns*
   Ness M.K., et al., incl. **Johnson J.W.**
   2026, MNRAS, 548, 637 – 656 arxiv:2605.20487

5. *Scylla at APOGEE: The Impact of Starbursts on the Chemical Evolution of the Magellanic Clouds*
   Escala I., et al., incl. **Johnson J.W.**
   2026, ApJ, 1003, 237 – 263 arxiv:2603.25909

6. *[C/N] Ages for Red Giants and their Implications for Galactic Archaeology*
   Roberts J.D., Pinsonneault M.H., Johnson J.A. Dubay L.O., **Johnson J.W.**
   2025, ApJ, 1002, 191 – 206 arxiv:2509.25321

7. *Challenges to the Two-Infall Scenario by Large Stellar Age Catalogs*
   Dubay L.O., Johnson J.A., **Johnson J.W.**, Roberts J.D.
   2025, ApJ, 1000, 244 – 262 arxiv:2508.00988

8. *The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*
   Otto J.M., et al., incl. **Johnson J.W.**
   2025, AJ, 171, 91 – 117 arxiv:2507.07264

9. *The Nineteenth Data Release of the Sloan Digital Sky Survey*
   SDSS Collaboration, et al., incl. **Johnson J.W.**
   2025, ApJS, 285, 9 – 47 arxiv:2507.07093

10. *Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy*
    Kollmeier J.A., et al., incl. **Johnson J.W.**
    2025, AJ, 171, 52 – 97 arxiv:2507.06989

11. *Many Elements Matter: Detailed Abundance Patterns Reveal Star-formation and Enrichment Differences among Milky Way Structural Components*
    Griffith E.J., Hogg D.W., Hasselquist S., **Johnson J.W.**, Price-Whelan A., Sit T., Stone-Martinez A., Weinberg D.H.
    2024, ApJ, 169, 280 – 297 arxiv:2410.22121

12. *Modeling the Galactic Chemical Evolution of Helium*
    Weller M.K., Weinberg D.H., **Johnson J.W.**
    2024, MNRAS, 583, 1517 – 1534 arxiv:2404.08765

13. *Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution*
    Dubay L.O., Johnson J.A., **Johnson J.W.**
    2024, ApJ, 973, 55 – 80 arxiv:2404.08059

14. *The APO-K2 Catalog. II. Accurate Stellar Ages for Red Giant Branch Stars Across the Milky Way*
    Warfield J.T., et al., incl. **Johnson J.W.**
    2024, AJ, 167, 208 – 231 arxiv:2403.03249

15. *Nature vs. Nurture: Distinguishing effects from stellar processing and chemical evolution on carbon and nitrogen in red giant stars*

Roberts J.D., et al., incl. **Johnson J.W.**
2024, MNRAS, 530, 149 – 166 arxiv:2403.03249

16. *The Scale of Stellar Yields: Implications of the Measured Mean Iron Yield of Core Collapse Supernovae*
Weinberg D.H., Griffith E.J., **Johnson J.W.**, Thompson T.A.
2023, ApJ, 973, 122 – 136 arxiv:2309.05719

17. *Untangling the Sources of Abundance Dispersion in Low-Metallicity Stars*
Griffith E.J., Johnson J.A., Weinberg D.H., Ilyin I., **Johnson J.W.**, Rodriguez-Martinez R., Strassmeier K.G.
2022, ApJ, 944, 47 – 67 arxiv:2210.01821

18. *Birth of the Galactic Disk Revealed by the H3 Survey*
Conroy C., et al., incl. **Johnson J.W.**
2022, submitted to AAS Journals, under peer review arxiv:2204.02989

19. *Primordial Helium-3 Redux: The Helium Isotope Ratio of the Orion Nebula*
Cooke R.J., Noterdaeme P., **Johnson J.W.**, Pettini M., Welsh L., Peroux C., Murphy M.T., Weinberg D.H.
2022, ApJ, 932, 60 – 76 arxiv:2203.11256

20. *Residual Abundances in GALAH DR3: Implications for Nucleosynthesis and Identification of Unique Stellar Populations*
Griffith E.J., Weinberg D.H., Buder S., Johnson J.A., **Johnson J.W.**, Vincenzo F.
2021, ApJ, 931, 23 – 50 arxiv: 2110.06240

21. *Chemical Cartography with APOGEE: Mapping Disk Populations with a Two-Process Model and Residual Abundances*
Weinberg D.H., et al., incl. **Johnson J.W.**
2021, ApJS, 260, 32 – 77 arxiv:2108.08860

22. *CNO dredge-up in a sample of APOGEE/Kepler red giants: Tests of stellar models and galactic evolutionary trends of N/O and C/N*
Vincenzo F., et al., incl. **Johnson J.W.**
2021, submitted to MNRAS, under peer review arxiv:2106.03912

23. *The Impact of Black Hole Formation on Population-averaged Supernova Yields*
Griffith E.J., Sukhbold T., Weinberg D.H., Johnson J.A., **Johnson J.W.**, Vincenzo F.
2021, ApJ, 921, 73 – 94 arxiv:2103.09837

24. *Nucleosynthesis signatures of neutrino-driven winds from proto-neutron stars: a perspective from chemical evolution models*
Vincenzo F., Thompson T.A., Weinberg D.H., Griffith E.J., **Johnson J.W.**, Johnson J.A.
2021, MNRAS, 508, 3499 – 3507 arxiv:2102.04920

25. *The Similarity of Abundance Ratio Trends and Nucleosynthetic Patterns in the Milky Way Disk and Bulge*
Griffith E.J., et al., incl. **Johnson J.W.**
2021, ApJ, 909, 77 – 101 arxiv:2009.05063